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Title of chapter: Chapter 13 - Transparency

Summary:

This chapter emphasizes the importance of transparency in developing trustworthy and responsible machine learning systems. It asserts that transparency enables stakeholders such as developers, managers, and regulators to comprehend how and why an AI system makes decisions. Without it, even well-performing systems may not be trusted or acknowledged. The chapter proposes factsheets as an important technique for increasing transparency. Similar to nutrition labels on food, factsheets give a structured record of a model's purpose, data, performance, fairness, and limitations. They enable all stakeholders to understand the system's capabilities and constraints, guaranteeing accountability and informed decision-making throughout the machine learning lifecycle.

The chapter also tackles the problem of maintaining the provenance of facts, which means ensuring that the information contained in the factsheet is accurate, authentic, and unaltered. It recommends employing immutable ledgers, such as blockchain technology, to securely record these facts. Each record/block contains: the fact itself, the time it was recorded, and a cryptographic hash that connects it to the preceding block. This makes tampering nearly impossible since changing one block would need modifying all following blocks across all distributed copies. Blockchains, whether permissioned (private: restricted to authorized users with credentials) or permissionless (public: open to anyone, allowing public verification of facts), ensure that once information is recorded, it cannot be changed, providing a reliable record of the system's development. The chapter concludes by illustrating how transparent documentation and verified provenance can build stakeholder confidence, leading to fairer and more ethical deployment outcomes.

What are two points that you learned after reading this chapter?:

- I learned that factsheets play a vital role in boosting transparency by systematically documenting all aspects of a machine learning model, helping both internal and external stakeholders understand and trust the system.
- I learned that blockchain technology can be used to maintain the provenance of facts by ensuring data integrity and preventing tampering, which strengthens accountability in AI system development.

What are two questions that you have about this chapter?:

- How can organizations balance the computational and communication costs of using blockchain for provenance tracking with the need for real-time updates in machine learning systems?
- What standards or regulatory frameworks currently exist (or should exist) to ensure consistency and reliability in how factsheets are created and maintained across different industries?

Do you have some initial thoughts or insights to approach the questions?:

- One possible approach is to explore hybrid models that use blockchain selectively. For example, only recording key checkpoints or cryptographic hashes of model states instead of entire datasets. This could reduce computational overhead while still ensuring integrity.
- I think it would be helpful to research existing AI governance frameworks, such as the IEEE standards on transparency, to see how they define documentation requirements. Developing universal templates or tools for factsheets could promote standardization and make compliance easier across organizations.